

Checklist for a DAQ system

Sensor. It is very important to know your signal and transducer well. "A lot of people make mistake in selecting the sensor, leading to junk acquisition that soils the entire system," says Nair. You need to choose a sensor and transducer that will give you the most accurate and reliable results.

Signal-conditioning requirement. Every signal that is acquired will show variation, so the signal-conditioning requirement will also vary. Having good knowledge of signal conditioning and getting proper consultation is very important. There are signal-conditioning systems that can be configured as per the requirement.

ADC. Selecting the right type of analogue-to-digital converter (ADC) is important as it will reflect the sampling rate and the data transfer rate. If you use a very high-resolution ADC, the number of codes will be high and it will require more memory to store.

Interface. As the signal acquired gets transferred to the storage drive or PC for further analysis, depending on the requirement of data transfer rate and sampling rate you can select the bus through which you want to communicate.

Drivers. It's the drivers that make your system futureproof. If the driver is proprietary and you can use it for certain time only, your DAQ system may not be useful later. A good driver allows you to bridge the gap between the application hardware and software.

Application software. "Application software allows the user to set up different channels, display measured values in real time, store data in files on the hard disk, generate test reports and view previously stored values," explains Gupta from AG Measurements.

paper roll ran up to 300 sheets for a 30-minute recording on an average. This also made analysis of the signal difficult as you had to manually view and measure all the signal levels. The floor space requirement was around 50 sq.m," shares Dr Shantilal Dhadiwal, a renowned psychiatrist. "With digital EEG, the signals are acquired directly into the desktop computer or a laptop and the storage is also digital. The software helps in zooming in and zooming out the anomalies in the signal giving better view for diagnosis. Also, the record, which is digital, can be shared (via CD or the Internet) with any other doctor sitting abroad."

Intelligent DAQ with FPGA

Data acquisition systems have become intelligent enough to make some of the decisions. "DAQs with built-in FPGAs are used for on-board measurements and control applications," informs Reddy.

"The intelligent DAQs with FPGA allows you to program the FPGA to take immediate action without transferring the data to the controller. The data acquired is analysed on the local memory with a latency of 1-2 seconds and then you can push the selective useful data to your controller," says Nair.

He explains, "If there is spike in the system that you are monitoring, you can send an immediate control signal like 'shut down' from your programmed FPGA and just send the interesting data, maybe the 'time' when the spike occurred, to your PC."

The intelligent DAQ system delivers high performance, user-configurable time and synchronisation as well as on-board decision-making. It gives you the flexibility to configure the hardware for simulation, bit-error-rate testing, flexible triggering and other applications that require precise timing and control.

Virtual instrumentation and modelling

Application software defines the usefulness of the data that is acquired by your hardware. It allows you to do useful analysis with the data acquired and send back the control signal for your system. The analysis can be done online, where signals are acquired and analysed immediately to generate the control signal, or off-line, where you analyse the data later to study and project trends. Certain DAQ software allow you to import and export data to common databases like Microsoft Excel and Access. Many DAQ hardware providers have their own software mod-

ules to present the useful results.

Another part in data acquisition where software plays a major role is modelling and simulation. "Modelling and simulation is very important in technical computing work-flow and acquiring data for computation becomes critical part of it," says Prashant Rao, technical manager, Mathworks India. "It helps you (the user) to figure out what exactly you would like to do with the data (for example, a feedback signal to a control system) in optimal manner before moving to the real product."

"Graphical programming has become the most popular feature of the software today. With this, you can click and get the data into the system without having to write and run the lengthy code," shares Rao.

Companies have developed software applications that allow you to build the entire data acquisition system without using the hardware. "Virtual instrumentation allows you to use highly interactive software along with modular, high-performance hardware to create a powerful computer-based instrumentation solution," shares Nair.

Beat the challenge

"The biggest challenge for DAQ is the growth in the number of integration points and short cycle time of deployment. Increase in functionalities due to newer regulations in safety, emissions, environment sensibility and drive to reduce the carbon footprint leads to complexity," informs Patwardhan.

"With varied applications of data acquisition, the requirement and challenges both are different. As we deal with data, purity of data, compression and consolidation are challenging areas," says Neelam Kumar, executive director, Aplab.

But with technological developments, all these challenges are manageable. Data acquisition systems are able to integrate and benefit from the developments in sensor, high-speed digitisation, bus and PC technologies. ■

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